

Data Science Practicum II Proposal: Youtube Thumbnail Virality Prediction Model

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I. ABSTRACT:

This practicum proposal focuses on the design and development of a predictive modeling system to assess the virality of a thumbnail. Thumbnails act as the critical determinant for clicks and overall engagement. In this project, we dive deep to see if the type of thumbnail determines the success rate or, in this case, high interaction or view count. This project builds a pipeline that allows users to input a thumbnail and receive a success rate calculation. The system uses historical data alongside CLIP and manually selected features to develop an XGBoost model to identify patterns between thumbnail images. The project presents a functioning prototype that allows users to select a long-form or short-form thumbnail option and upload a thumbnail to generate a success rate for the input, as well as providing some improvement recommendations.

Index Terms

CCLIP, Computer Vision, Predictive Modeling, Social Media Analytics, Machine Learning, Sentiment, Feature Extraction, PCA, Video Virality, click-through rate.

II. INTRODUCTION/BACKGROUND:

With the assistance of AI, as well as many creative marketers and content creators, thumbnails on YouTube are the first thing users see as an introduction to the video they might view. Globally, over 3.37 billion internet users consume video content, yet the sheer volume of uploads makes standing out increasingly difficult [1].

A study by Vydia found that 65% of viewers are more likely to watch a video based on the thumbnail, while the remaining percentage is influenced by the title [2]. The average click-through rate for videos with optimized thumbnails is between 12%–25%. The increase in viewer engagement when using personalized thumbnails is 25%–30%. The channel growth rate for creators who regularly choose to optimize their thumbnails is 20%–25% higher than those who don't [2].

Recent research underscores the complexity of this visual decision-making process. According to Cui et al. [3], the "thumbnail" which is comprised of the title, an image, and captions is the primary factor in a viewer deciding whether or not to watch. Researchers' analysis of over 16,000 YouTube covers reveals that specific "sentiment markers," such as strong emotional expressions in thumbnails (both positive and negative), significantly lead to more views. However, there can also be additional factors in YouTube covers that make users click. This leaves a gap. Currently, while many creators spend significant time on

thumbnail design, 48% of content marketers feel they are not utilizing video content to its full potential due to a lack of technical skills in analyzing these variables [1].

This explains that although there are many diverse creative tools, there is still room for growth in predictive visual analytics. There are online websites like Social Blade created to help understand analytical growth [4] by providing live datasets and insights on creators; however, they still miss the mark by not recommending ways to create a better thumbnail or comparing creators' thumbnails to other creators' thumbnails to understand their success. A system that integrates computer vision and historical data analysis can address these limitations and assist content creators.

III. PROBLEM STATEMENT:

This research seeks to identify the visual characteristics of YouTube thumbnails that correlate with high user engagement and success rates. Furthermore, it aims to develop a predictive model capable of estimating the potential performance of a video thumbnail based on selected features. Virality in this project will be defined as having a high view count.

IV. RELATED WORK:

Previous research has explored identifying YouTube thumbnail success. Early work focused on thumbnail selection through classical feature selection techniques [5][6]. Other researchers have utilized approaches such as convolutional neural networks (CNNs) for selecting video thumbnails. While focusing on selecting a thumbnail from a successful video, this research first evaluates the success of a video before creating the thumbnail [6]. Although this research addresses the broader success of a YouTube video, it still fails to address the challenge of creating a thumbnail when video success metrics are unknown.

Empirical investigations indicate that while general image recognition is a mature technology, the spatial and emotional understanding of how thumbnail sentiments interact with titles and captions remains an active area of improvement for predictive algorithms [3].

Some significant modern works focused on improving thumbnail success include the A/B testing features provided by YouTube. This process allows creators to test up to three different thumbnail variations. Creators can choose to test titles only, thumbnails only, or both titles and thumbnails simultaneously. This enables creators to monitor analytics for different styles, helping them grow their channels and design future thumbnails based on high-performing layouts [7].

With the introduction of AI in recent years, programs like TubeBuddy AI have emerged to help identify the most effective thumbnails. TubeBuddy AI is trained on human behavior to pinpoint where an audience's eyes will land [8]. It utilizes heatmaps on thumbnails to allow users to visualize exactly what their audience sees.

Through this research to develop a predictive model, we address the understanding of behavioral semantics on thumbnails while incorporating additional thumbnail data to make valid predictive decisions, instead of only comparing metrics to a single creator's success.

V. METHODOLOGY/APPROACH

The methodology/approach of building this predictive model is as follows:

A. Data Collection

Data was collected from the YouTube Data API v3. Due to request limits, data from the categories *People & Blogs*, *Gaming*, and *Music* were prioritized. Specific API functions and methods used include:

- `YouTube.search.list`: Returns a list of video IDs from search queries.
- `YouTube.videos.list`: Returns statistical information such as title, view count, and duration.
- `Request.get()`: Retrieves the actual image file of the thumbnail.

Data was pulled using categorical queries such as: "*official music video 2024*", "*gaming walkthrough gameplay 2024*", and "*day in my life vlog 2023*". The resulting dataset includes information such as `video_id`, `view_count`, `is_short`, `tags`, and `thumbnail_path` and more.

B. Data Preprocessing

During this phase, two primary issues were resolved. First, view count was transformed into a logarithmic format to handle the large distribution gap (ranging from single digits to over one billion). However this was only used for EDA purposes and was not pushed into the model.

Second, to address temporal bias where older videos naturally have more views than recent ones a normalization formula was applied to calculate daily engagement:

$$views_per_day = \frac{view_count}{age_days} \quad (1)$$

The variable `views_per_day` was then log-transformed to `log_views_per_day` before being utilized as the model's target variable.

C. Feature Selection

Two feature extraction methods were utilized to select inputs for the model:

- 1) **CLIP Embeddings**: CLIP (Contrastive Language-Image Pre-training) is designed to understand relationship between image and text [9]. However it can be used only with images by being paired with features [10]. In this project CLIP was used to capture high-level semantic information from each thumbnail, producing a 512-dimensional encoded fingerprint. To optimize these features for the XGBoost model, Principal Component Analysis (PCA) was applied to reduce the dimensionality to 50, which successfully retained 58.44% of the original variance.
- 2) **Manually Selected Features**: A total of 15 features were initially extracted to provide the model with specific design metrics (As shown in figure 1).
 - **Full Feature Set**: The initial extraction included: *brightness*, *contrast*, *saturation*, *colorfulness*, *edge density*, *sharpness*, *avg_red*, *avg_green*, *avg_blue*, *warm ratio*, *dark ratio*, *bright ratio*, *aspect ratio*, *face count*, and *face presence*.
 - **Feature Selection Results**: After conducting a feature importance analysis against `log_views_per_day` using a significance threshold of $\geq +0.05$ or ≤ -0.05 , the following 8 features were retained as the most impactful:
Contrast, *Saturation*, *Colorfulness*, *Edge Density*, *Sharpness*, *Warm Ratio*, *Dark Ratio*, and *Face Count*.

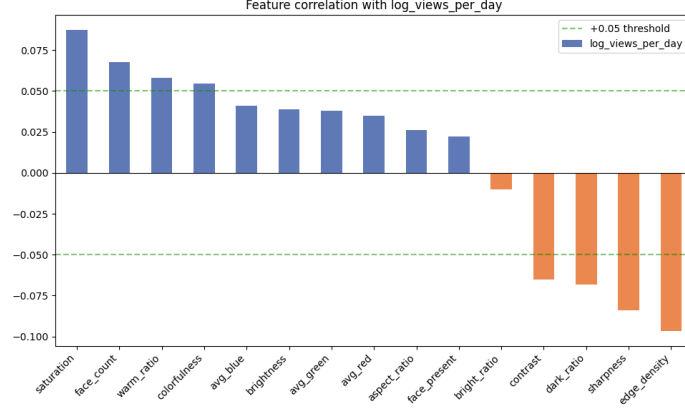


Fig. 1. feature selection graph

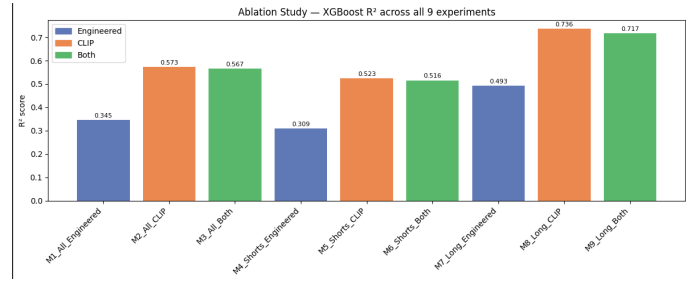


Fig. 2. R^2 score for 9 XGBOOST models

VI. RESULT

A. Model Selection and Experimental Design

XGBoost was selected as the primary model for this project due to its proficiency in handling engineered features and its robust predictive capabilities using gradient boosting ensemble techniques.

An ablation study, comprising nine distinct experiments, was conducted to evaluate the two feature extraction methods and the performance differences between short-form and long-form video thumbnails. The data split for the XGBoost model followed an 80/20 train-test ratio, with R^2 serving as the primary metric for performance evaluation.

1) *Baseline Comparison*: A Ridge regression model was established as a baseline to define a minimum performance threshold. Ridge regression assumes a global linear relationship across all video types, which proved insufficient for this dataset. The baseline model achieved an R^2 score of only 0.045, explaining just 4.5% of the variance in thumbnail virality.

2) *Ablation Study (M1–M9) (fig 2)*: As detailed in Table I, the experiments were structured as follows:

- **Full Dataset (M1–M3)**: These models utilized the complete dataset (short and long-form).
 - M1: Manually selected features only.
 - M2: CLIP embeddings only.
 - M3: Combined CLIP and selected features.
- **Short-Form Subsets (M4–M6)**: These models focused exclusively on short-form thumbnails.
 - M4: Manually selected features.
 - M5: CLIP embeddings.

- M6: Combined features.
- **Long-Form Subsets (M7–M9):** These models analyzed only long-form thumbnails.
 - M7: Manually selected features.
 - M8: CLIP embeddings.
 - M9: Combined features.

TABLE I
SUMMARY OF THE NINE ABLATION EXPERIMENTS

Model	Data Subset	Features	R^2
M1	All	Selected	0.345
M2	All	CLIP	0.573
M3	All	CLIP + Selected	0.567
M4	Short form	Selected	0.309
M5	Short form	CLIP	0.523
M6	Short form	CLIP + Selected	0.516
M7	Long form	Selected	0.493
M8	Long form	CLIP	0.736
M9	Long form	CLIP + Selected	0.717

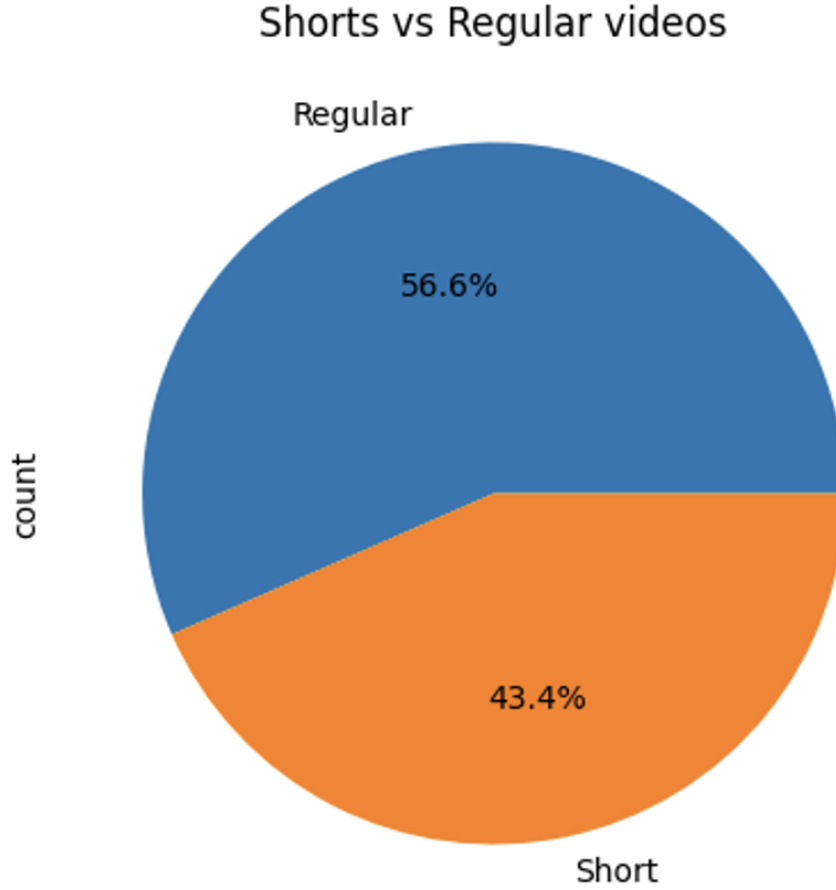


Fig. 3.

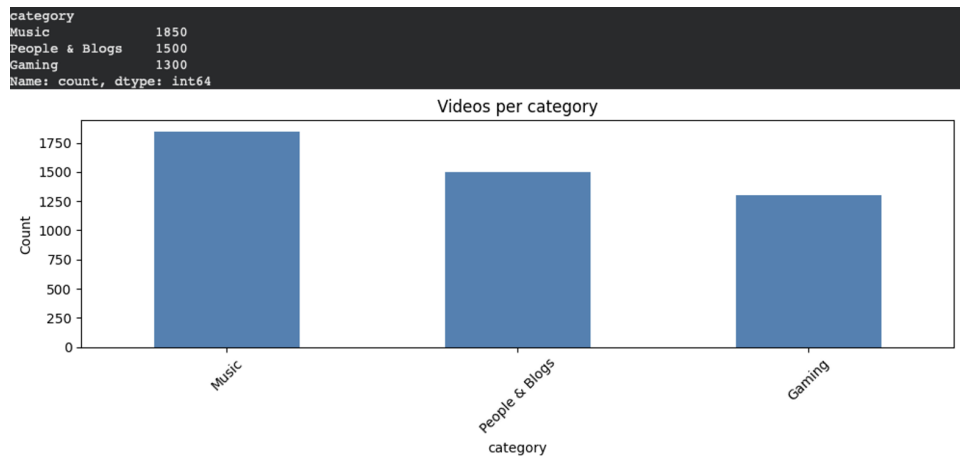


Fig. 4.

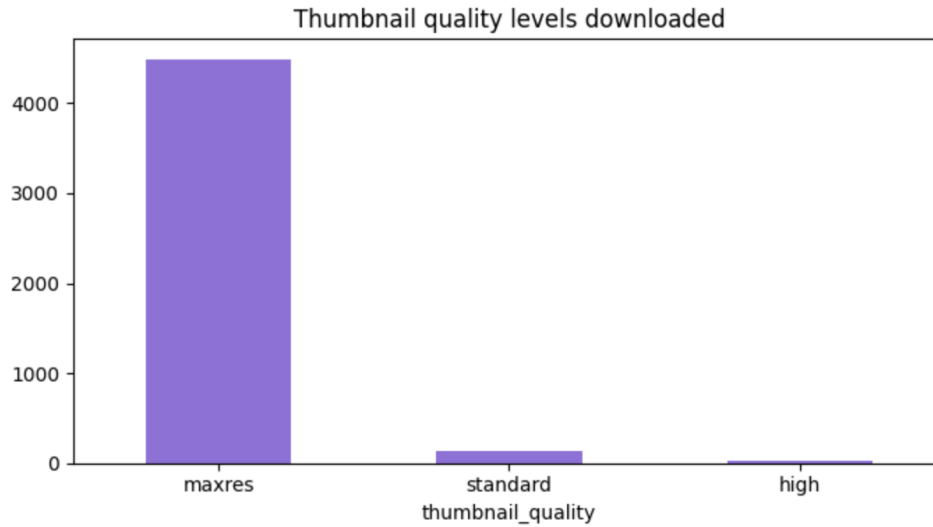


Fig. 5.

VII. DATA ANALYSIS

The data we are working with includes thumbnail images/jpg files from shorts and long-form videos. In total, the dataset used is 4,650. 56.6% of the data are from long-form video thumbnails, while the remaining 43.4% are short-form video thumbnails (Figure: 3).

The data collected spans three categories: Music, People & Blogs, and Gaming. Of the 4,650 entries, 1,850 are Music thumbnails, 1,500 are People & Blogs thumbnails, and 1,300 are from Gaming video thumbnails (Figure: 4). The majority of the thumbnail dataset is of max resolution, and only a small amount are standard and high (Figure: 5).

VIII. KEY FINDINGS & INSIGHTS:

Findings in this project demonstrate that XGBoost outperforms Ridge regression by a significant margin, as illustrated in Figure 6. The plotted graph shows that XGBoost predictions align closely with the black dashed line, representing an ideal prediction. In contrast, the Ridge regression predictions result in a

horizontal distribution across a single prediction value, indicating a failure to capture the variance in the dataset.

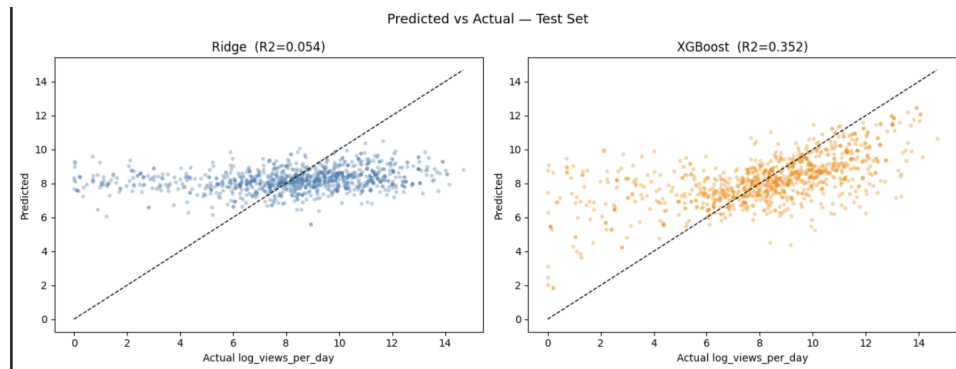


Fig. 6.

To interpret the XGBoost model and determine the influence of manually selected features, SHAP (Shapley Additive explanations) values were calculated for the eight primary features. The analysis reveals that `dark_ratio` and `edge_density` are the most significant features impacting predictions specifically acting as indicators of low-performing thumbnails. By contrast, `saturation`, `warm_ratio`, `contrast`, `face_count`, and `colorfulness` were found to have a positive impact on the predicted success rate.

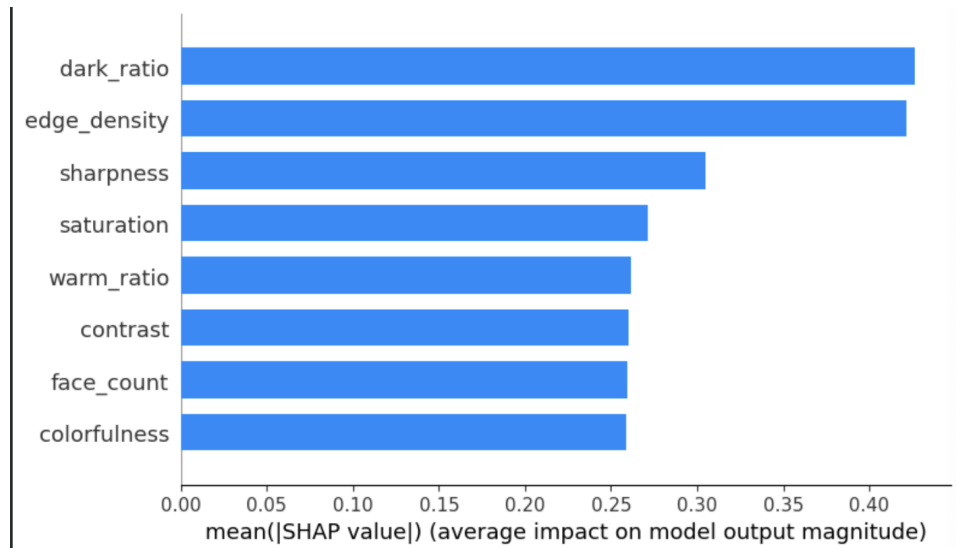


Fig. 7.

When focusing on the dataset used for training, Figure 8 illustrates the visual distinctions the model identifies between high-performing and low-performing thumbnail designs for both short-form and long-form video formats.

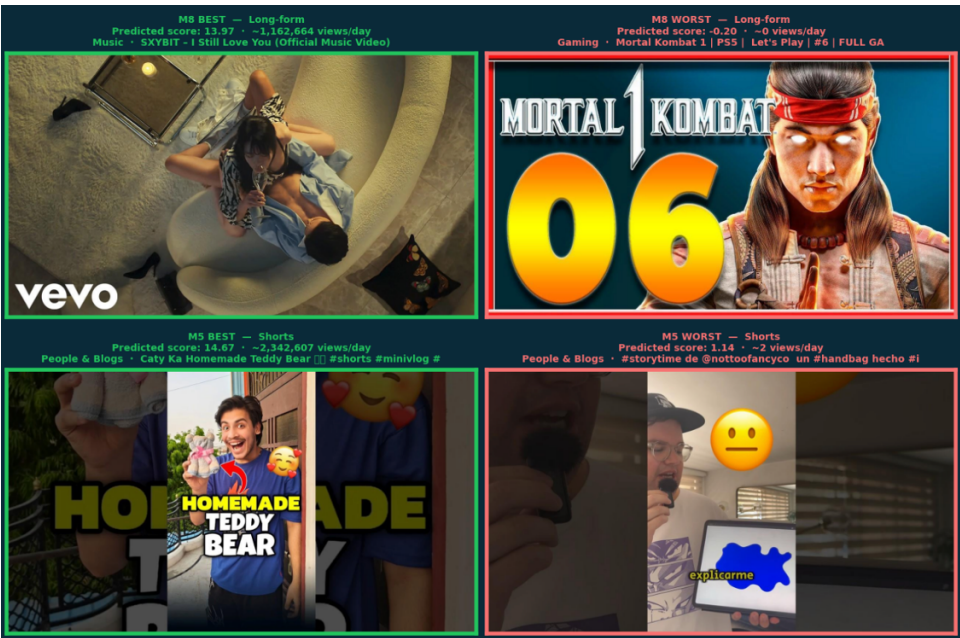


Fig. 8.

With nine different experimental models processed, model **M8** was selected for predicting thumbnail virality for long-form videos, and model **M5** was selected for predicting virality for short-form content due to the high R^2 scores of $M8=0.736$ and $M5=0.523$.

A. Model Implementation

When interacting with the model the user interface requires the selection of either short-form or long-form video types prior to generating a prediction. When testing the model against unseen thumbnails, as illustrated in Figure 9, the system correctly predicted the virality of the "BISKIDS songs" YouTube Short thumbnail, aligning with the video’s actual performance metrics.

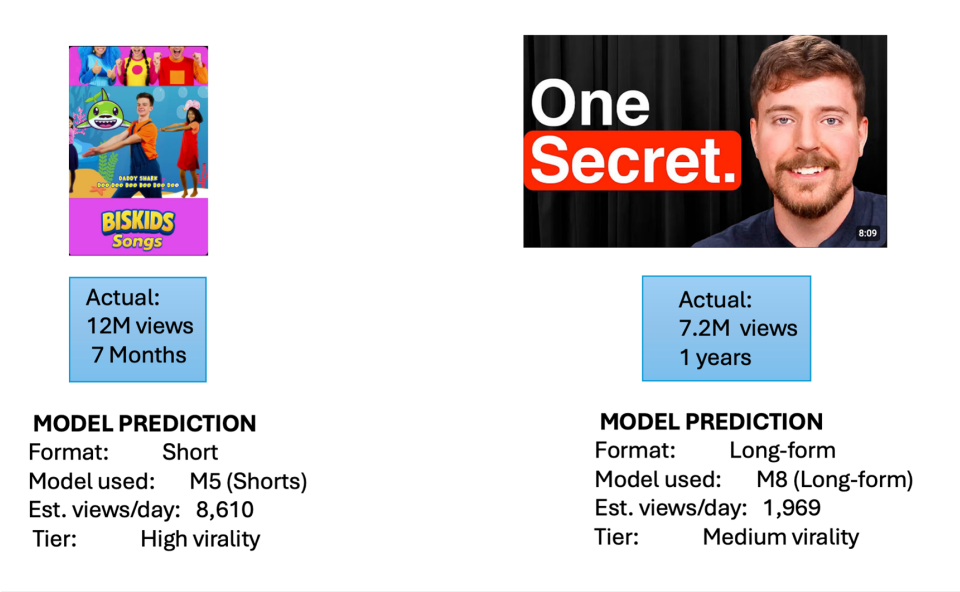


Fig. 9.

In a comparative test using a long-form video thumbnail, the model predicted "medium virality." When compared to the YouTube Short performance, the model's predictive logic remains consistent with empirical data: the "BISKIDS" video accumulated a higher view velocity over a two-month period than the "MrBeast One Secret" video, which recorded 7.2 million views over the span of a full year. This suggests the model successfully distinguishes virality expectation.

IX. CONCLUSION AND FUTURE WORK

A. Conclusion

The developed model demonstrates a strong ability to identify visual features that distinguish between high-performing and low-performing thumbnail designs. However, significant opportunities for improving accuracy remain. A primary limitation of this research was the lack of data focused on the historical growth of view counts for specific thumbnails. Access to such time-series data would have allowed for a deeper understanding of the specific trends and "tipping points" that lead to virality.

Based on the final validation, this model serves as a viable real-world tool for assisting with design decisions within the *Music*, *People & Blogs*, and *Gaming* categories when used alongside other content optimization tools. However, full dependency on this model is currently not recommended due to the limited scope and size of the training dataset.

B. Future Work

There are several avenues through which this research can be expanded to enhance its predictive power and functionality:

- **Developing a More Diverse Dataset:** The current model relies on hard-coded queries for specific categories. Future iterations should involve a more robust data ingestion pipeline that covers a broader array of categories and different search queries. This would enable the model to develop a generalized understanding of visual designs across the entire platform rather than remaining limited to specific niches.
- **Expanding User Input Variables:** While the current system differentiates between short-form and long-form content, incorporating additional user-provided variables, such as intended video duration and specific category selection, would allow for more precise benchmarking. This would enable the system to evaluate an uploaded thumbnail against its direct competitors within a specific field rather than comparing it against unrelated content.
- **Personalized Channel Analysis:** As suggested by Dr. Kellen Sorauf, refining the model to be creator-specific would significantly improve performance. By allowing users to integrate their own historical YouTube analytics (e.g., previous thumbnails, click-through rates, and subscriber counts), the model could leverage both platform-wide trends and the specific behavior of a creator's unique audience to provide personalized design recommendations.

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